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shaping futures

CS201 – Data Structures II

Recap of Skiplists Analysis

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Analysis

- A random variable X is a function from the **sample space** to the set of **real numbers**
- Example:
 - X denotes the number of heads appearing when throwing three coins
 - The random variable X takes on one of the values 0, 1, 2, and 3 meaning it has 0, 1, 2, or 3 heads with probabilities
 - $\Pr[X = 0] = \Pr[\{\text{TTT}\}] = \frac{1}{8},$
 - $\Pr[X = 1] = \Pr[\{\text{HTT, THT, TTH}\}] = \frac{3}{8},$
 - $\Pr[X = 2] = \Pr[\{\text{HHT, HTH, THH}\}] = \frac{3}{8},$
 - $\Pr[X = 3] = \Pr[\{\text{HHH}\}] = \frac{1}{8}.$

Analysis

- The expected value of a (discrete) random variable X with range S is defined as

$$E[X] = \sum_{x \in S} x \cdot \Pr[X = x]$$

- Example:

– Let X denote the number appearing when throwing a die, then the expected value of X is:

$$E[X] = \sum_{k=1}^6 k \cdot \Pr[X = k] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = \frac{7}{2}$$

- $E[X]$ represents the **mean** of the random variable X
- Important property: Linearity of expectation that is: $E[X + Y] = E[X] + E[Y]$, more generally:

$$E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i]$$

Analysis

- Indicator random variables are binary variables which are useful when we want to count something
- Example:
 - We toss a fair coin k times and we want to know the expected number of times the coin turns up as heads
 - Using expected value:

$$E[X] = \sum_{i=1}^k i \cdot \Pr[X = i]$$

$$= \sum_{i=1}^k i \cdot \binom{k}{i} / 2^k$$

$$= \sum_{i=1}^k k \cdot \binom{k-1}{i-1} / 2^k$$

using binomial identity $i \cdot \binom{k}{i} = k \cdot \binom{k-1}{i-1}$

$$= k \cdot \sum_{i=1}^k \binom{k-1}{i-1} / 2^k$$

$$= k \cdot 2^{(k-1)} / 2^k$$

$$= k/2$$

using binomial identity $\sum_{i=0}^{k-1} \binom{k-1}{i} = 2^{k-1}$

Analysis

- We can also use an indicator random variable like so:

$$I_i = \begin{cases} 1 & \text{if } i^{\text{th}} \text{ coin comes as head} \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Then, } E[I_i] = \frac{1}{2} * 1 + \frac{1}{2} * 0 = \frac{1}{2}$$

$$\text{For } X = \sum_{i=1}^k I_i,$$

$$\begin{aligned} E[X] &= E\left[\sum_{i=1}^k I_i\right] \\ &= \sum_{i=1}^k E[I_i] \\ &= \sum_{i=1}^k \frac{1}{2} \\ &= \frac{k}{2} \end{aligned}$$

Analysis

- Expected value of an indicator random variable I_A for an event A is just the probability of that event A .

$$E[I_A] = \Pr(A)$$

- **Problem:** Suppose there is a dinner party where n men check their hats. The hats are mixed up during dinner, so that afterward each man receives a **random** hat. In particular, each man gets his own hat with probability $\frac{1}{n}$. What is the expected number of men who get their own hat?

Solution

- Let p be the probability of each person getting his hat $= \frac{1}{n}$
- We define an indicator random variable I_i as follows:

$$I_i = \begin{cases} 1 & \text{if } i^{\text{th}} \text{ man gets his hat} \\ 0 & \text{otherwise} \end{cases}$$

$$E[I_i] = E[\Pr(I_i = 1)] = \frac{1}{n}$$

The number of men who get their own hat is

$$E\left[\sum_{i=1}^n I_i\right] = \sum_{i=1}^n E[I_i] = \sum_{i=1}^n \frac{1}{n} = 1$$

Problem

- Suppose that we independently flip n **biased** coins, each with probability p of coming up heads. What is the expected number of heads?
 - We can use the binomial probability formula which says:
 - $\Pr(k \text{ successes in } n \text{ trials}) = \binom{n}{k} p^k q^{n-k}$
 - Where,
 - n = number of trials
 - k = number of successes
 - $n - k$ = number of failures
 - p = probability of success in one trial
 - $q = 1 - p$ = probability of failure in one trial
 - Probability $\Pr[X = k] = \binom{n}{k} p^k q^{n-k}$

Analysis

- Using expected value:

$$E[X] = \sum_{k=1}^n k \cdot \Pr[X = k]$$
$$= \sum_{k=1}^n k \cdot \binom{n}{k} p^k q^{n-k}$$

$$= \sum_{k=1}^n n \cdot \binom{n-1}{k-1} p^k q^{n-k}$$

using binomial identity $i \cdot \binom{k}{i} = k \cdot \binom{k-1}{i-1}$

$$= np \sum_{k=1}^n \binom{n-1}{k-1} p^{k-1} q^{n-k}$$

$$= np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j q^{n-j-1}$$

using $j=k-1$

$$= np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j q^{n-j}$$

Analysis

- Using binomial theorem

$$\sum_{j=0}^{n-1} \binom{n-1}{j} p^j q^{n-1-j} = (p+q)^{n-1}$$
$$= np(p+q)^{n-1}$$

- Since

$$-(p+q) = 1 \Rightarrow (p+q)^{n-1} = 1$$

$$= np$$

Skiplist Analysis

- **Lemma 4.2.:** Let T be the number of times a fair coin is tossed up to and including the first time the coin comes up heads. Then $E[T] = 2$.
- Proof:
 - We stop tossing the coin when we first get a head
 - We define an indicator variable I_i which is set to 0 (if coin is tossed $< i$ times) and 1 otherwise
 - $I_i = \begin{cases} 0 & \text{if the coin is tossed less than } i \text{ times} \\ 1 & \text{if the coin is tossed } i \text{ or more times} \end{cases}$
 - $I_i = 1$ if and only if the first $i - 1$ coin tosses are tails

$$E(I_i) = \Pr(I_i = 1) = \frac{1}{2^{i-1}}$$

$$\text{Total coin tosses } T = \sum_{i=1}^{\infty} I_i$$

Skiplist Analysis – Lemma 4.2 proof

$$\begin{aligned} E[T] &= E \left[\sum_{i=1}^{\infty} I_i \right] \\ &= \sum_{i=1}^{\infty} E[I_i] \\ &= \sum_{i=1}^{\infty} 1/2^{i-1} \\ &= 1 + 1/2 + 1/4 + 1/8 + \dots \\ &= 2 . \end{aligned}$$

Skiplist Analysis

- **Lemma 4.3.:** The expected number of nodes in a skiplist containing n elements, not including occurrences of the sentinel, is $2n$
- Proof:
 - A node in skiplist is selected at level L_k depends on the outcome of a fair coin toss coming as head
 - Probability of selection of a node at L_0 is $p_0=1$ as all nodes are always included in L_0
 - For level L_1 , $p_1=1/2$ (subjected to the fair coin toss being head)
 - For level L_2 , $p_2=1/2 * p_1=1/2^2$ since the nodes from L_1 will be used at level L_2
 - For level L_r , $p_r=1/2^r$
- Expected number of nodes at level $L_r = n/2^r$
- Expected number of nodes at all levels = $\sum_{i=0}^{\infty} n/2^i = n \left(\sum_{i=0}^{\infty} \frac{1}{2^i} \right) = 2n$

Skiplist Analysis

- **Lemma 4.4.:** The expected height of a skiplist containing n elements is at most $\log n + 2$
- Proof:
 - Let r be the level of skiplist: $r = \{1, 2, 3, \dots, \infty\}$
 - We define an indicator random variable: $I_r = \begin{cases} 0 & \text{if } L_r \text{ is empty} \\ 1 & \text{if } L_r \text{ is not empty} \end{cases}$
 - Height of the skiplist: $h = \sum_{r=0}^{\infty} I_r$
 - I_r is never more than the length, $|L_r|$, of L_r :

$$E[I_r] \leq E[|L_r|] = n/2^r$$

Skiplist Analysis – Lemma 4.4 proof

$$\begin{aligned}
 E[h] &= E\left[\sum_{r=1}^{\infty} I_r\right] \\
 &= \sum_{r=1}^{\infty} E[I_r] \\
 &= \sum_{r=1}^{\lfloor \log n \rfloor} E[I_r] + \sum_{r=\lfloor \log n \rfloor + 1}^{\infty} E[I_r] \\
 &\leq \sum_{r=1}^{\lfloor \log n \rfloor} 1 + \sum_{r=\lfloor \log n \rfloor + 1}^{\infty} n/2^r \\
 &\leq \log n + \sum_{r=0}^{\infty} 1/2^r \\
 &= \log n + 2 .
 \end{aligned}$$

The value of I_r cannot be great than 1 for all levels less than or equal to $\log n$, and for the rest, $E[I_r]$ can never be more than total number of nodes at level r i.e. $n/2^r$:

$$\begin{aligned}
 &\sum_{r=\log(n)+1}^{\infty} \frac{n}{2^r} \\
 &\sum_{r=0}^{\infty} \frac{n}{2^{r+\log(n)+1}} \\
 &\sum_{r=0}^{\infty} \frac{n}{2^r \cdot 2^{\log(n)} \cdot 2} \\
 &\sum_{r=0}^{\infty} \frac{n}{2^r \cdot n \cdot 2} \\
 &\sum_{r=0}^{\infty} \frac{1}{2^{r+1}} \\
 &\sum_{r=1}^{\infty} \frac{1}{2^r}
 \end{aligned}$$

Skiplist Analysis

- **Lemma 4.5.:** The expected number of nodes in a skiplist containing n elements, including all occurrences of the sentinel, is $2n + O(\log n)$.
- Proof:
 - The expected no. of nodes in a skiplist is $2n$ (see lemma 4.3)
 - There are always h sentinels in a skiplist of height h .
 - The expected height h of skiplist is $\log(n)+2$ (see lemma 4.4)
 - So expected number of nodes in a skiplist with n elements and all occurrences of sentinel will be $2n+\log(n)+2 = 2n+O(\log(n))$

Skiplist Analysis

- **Lemma 4.6.:** The expected length of the search path for any node, u , in L_0 is at most $2\log n + O(1) = O(\log n)$.
- Proof:
 - S is the search path length for node u in a skiplist with height h
 - Let S_r be the number of forward search path steps at level L_r
 - $E[S_r] \leq 1$ and $S_r \leq |L_r|$ as we can't take more steps than $|L_r|$

$$E[S_r] \leq E[|L_r|] = n/2^r$$

Skiplist Analysis – Lemma 4.6 proof

$$\begin{aligned} E[S] &= E\left[h + \sum_{r=0}^{\infty} S_r\right] \\ &= E[h] + \sum_{r=0}^{\infty} E[S_r] \\ &= E[h] + \sum_{r=0}^{\lfloor \log n \rfloor} E[S_r] + \sum_{r=\lfloor \log n \rfloor+1}^{\infty} E[S_r] \\ &\leq E[h] + \sum_{r=0}^{\lfloor \log n \rfloor} 1 + \sum_{r=\lfloor \log n \rfloor+1}^{\infty} n/2^r \\ &\leq E[h] + \sum_{r=0}^{\lfloor \log n \rfloor} 1 + \sum_{r=0}^{\infty} 1/2^r \\ &\leq E[h] + \sum_{r=0}^{\lfloor \log n \rfloor} 1 + \sum_{r=0}^{\infty} 1/2^r \\ &\leq E[h] + \log n + 3 \\ &\leq 2 \log n + 5 = O(\log(n)) \end{aligned}$$



Summary

- We discussed Skiplist and the two implementation SkiplistSet and SkiplistList
- We looked at their properties, structures and operations
- We looked at the complexity analysis of the Skiplist operations

Book Readings

- ODS: Chapter 4 pages: 87-102

